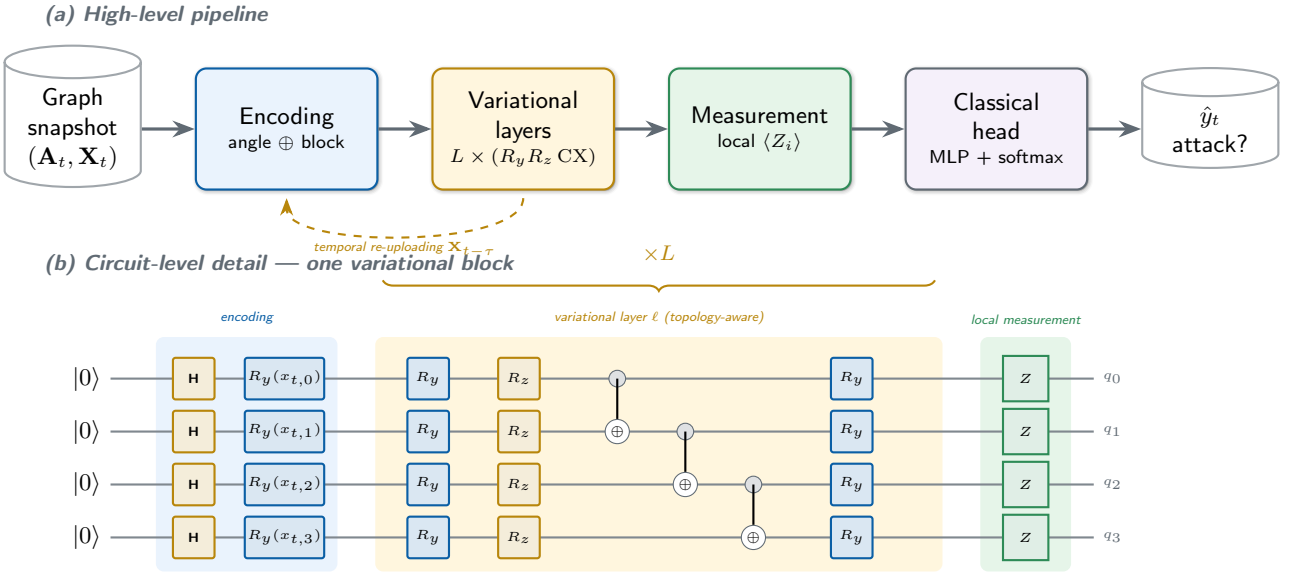
DynaQGNN-IDS — Proposed Ansatz Architecture



Encoding: angle $R_y(x_i)$ + Hadamard for amplitude spread; block-encoding of adjacency $\mathbf{A}_t$ via structured CX pattern.
Variational: $\boldsymbol{\theta}$, $\boldsymbol{\phi}$ learnable; CNOT topology follows edges of $\mathbf{A}_t$ $\rightarrow$ *topology-aware ansatz* (mitigates barren plateau via structured entanglement).
Measurement: *local* observables $\langle Z_i \rangle$ feed into classical MLP head $\rightarrow$ $\hat{y}_t$. Local cost avoids global barren plateau (Cerezo et al. 2021a).